

KAPREKAR CONTEST – FINAL – SUB-JUNIOR

(Classes VII and VIII)

AMTI – NOVEMBER, 2019.

- Instructions:
1. Answer all the questions.
 2. Elegant and novel solutions will get extra credits.
 3. Diagrams and explanations should be given wherever necessary.
 4. Fill in FACE SLIP and your rough working should be in the answer paper itself.
 5. Maximum time allowed is THREE hours.
 6. All questions carry equal marks.

1. Let a_n be the units place of $1^2 + 2^2 + 3^2 + \dots + n^2$. Prove that the decimal $0.a_1a_2a_3\dots a_n\dots$ is a rational number and represent it as $\frac{p}{q}$, where p and q are natural numbers.
2. (a) Find the positive integers m, n such that $\frac{1}{m} + \frac{1}{n} = \frac{3}{17}$
 (b) Find the positive integers m, n, p such that $\frac{1}{m} + \frac{1}{n} + \frac{1}{p} = \frac{3}{17}$.
 (c) Using this idea, prove that we can find for any positive integer k, k distinct integers, $n_1, n_2, \dots, n_k$ such that $\frac{1}{n_1} + \frac{1}{n_2} + \dots + \frac{1}{n_k} = \frac{3}{17}$.
3. Does there exist a positive integer which is a multiple of 2019 and whose sum of the digits is 2019? If no, prove it. If yes, give one such number.
4. In a triangle XYZ , the medians drawn through X and Y are perpendicular. Then show that XY is the smallest side of XYZ .
5. Let ΔPQR be a triangle of area 1 cm^2 .
 Extend QR to X such that $QR = RX$; RP to Y such that $RP = PY$ and PQ to Z such that $PQ = QZ$. Find the area of ΔXYZ .
6. Find the real numbers x and y given that $x - y = \frac{3}{2}$ and $x^4 + y^4 = \frac{2657}{16}$.
7. The difference of the eight digit number $ABCDEFGH$ and the eight digit number $GHEFCDAB$ is divisible by 481. Prove that $C = E$ and $D = F$.
8. $ABCD$ is a parallelogram with area 36 cm^2 . O is the intersection point of the diagonals of the parallelogram. M is a point on DC . The intersection point of AM and BD is E and the intersection point of BM and AC is F . The sum of the areas of the triangles AED and BFC is 12 cm^2 . What is the area of the quadrilateral $EOFM$?

